WIRELESS:

Today, Wireless LANs (WLAN) are used in most enterprises. Most devices now have wireless capabilities. Wireless enhancing the mobility and seamless connectivity.

Access point is a wireless device. That connect multiple wireless devices, provides seamless connectivity to users.

If your network having 50,60 APs, configuration cab be time consuming. To manage multiple APs,
WLC (Wireless LAN controller) fix this problem. WLC manage multiple APs.

Wireless Technologies-

1. WPAN (wireless personal area)
2. WLAN (wireless local area network)
3. WMAN (wireless metropolitan area network)

WIRELESS Technologies:

WPAN- A personal area (PAN) is a network that exist a relatively small area connect electronic devices such as **desktop**, **computer**, **printer**, **scanner**, **and notebook**, these devices use **Bluetooth** to connect to WPAN. Range of WPAN **>5-10 meters.**

WLAN- A WLAN network connects multiple wireless devices over a local area network. It connects wireless devices to backbone area (LAN), using **AP (Access Point)**. It spreads approximately **100 meter (328 feet).**

WMAN- AWAN network is wireless communication network that covers a large geographic area, such as a city, suburb. Wireless signal provides point to **point or point** to **multipoint**

Point to point- A point to point (PtP) wireless connection are used to connect two locations together using directional antennas with Los **(line of sight)**.

Point to multipoint- Point to multi point have large number of nodes, end destination and end users.

Wireless network usually consists of these devices-

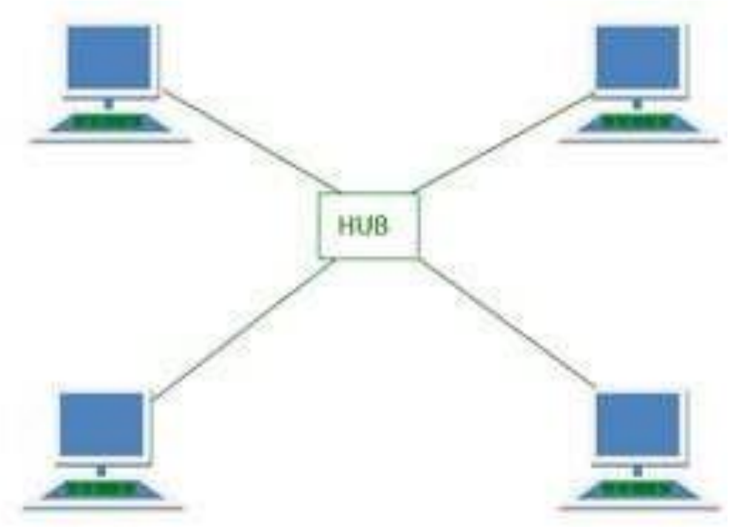
- Client with wireless adaptor
- Access point
- Cisco WLC device, centralized device that configuration of APs.
- The Standard we use for wireless LANs are Define by IEEE (International Electrical Electronic and engineers)

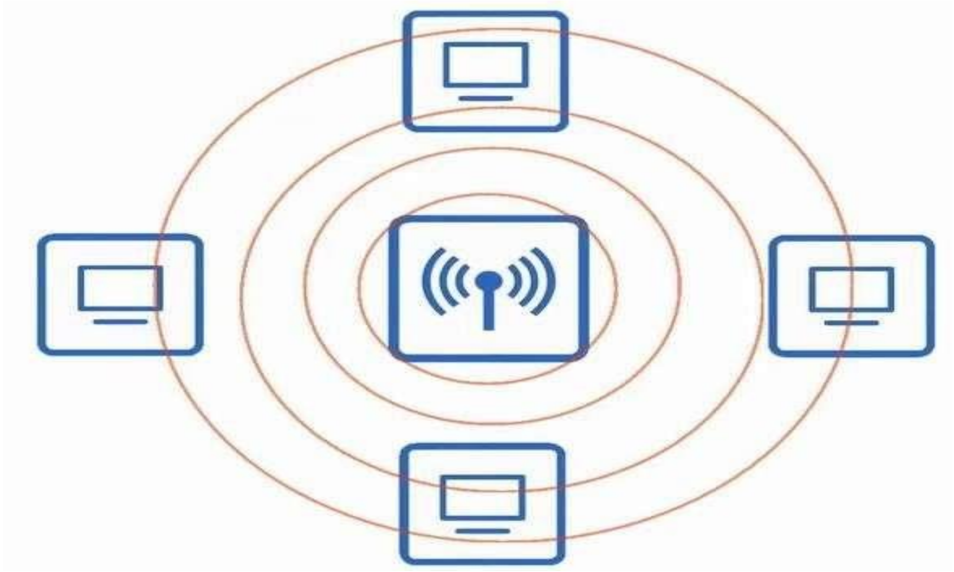
802.11

- 802.11 standard provides interoperability with other vendor Devices.

Wireless Devices has some Issues-

- All Devices within range receive all frames just like Ethernet HUB.
- Privacy of data within the LAN is greater concern.
- CSMA/CA (Carrier sense multi access collision advance) is used to facilitate Half Duplex communication.





Radio Frequency-

Frequency- Measure the number of UP/DOWN cycles per a unite of time.

The most common measurement of frequency is Hertz

1. Hz (Hertz)= Cycles Per Second.
2. kHz (Kilohertz)= 1,000 cycle per seconds
3. Mhz (Megahertz)= 1,000,000 cycle per seconds
4. Ghz (GigaHertz)= 1,000,000,000 cycle per seconds
5. THz (TeraHertz)= 1,000,000,000,000 Cycle per seconds.

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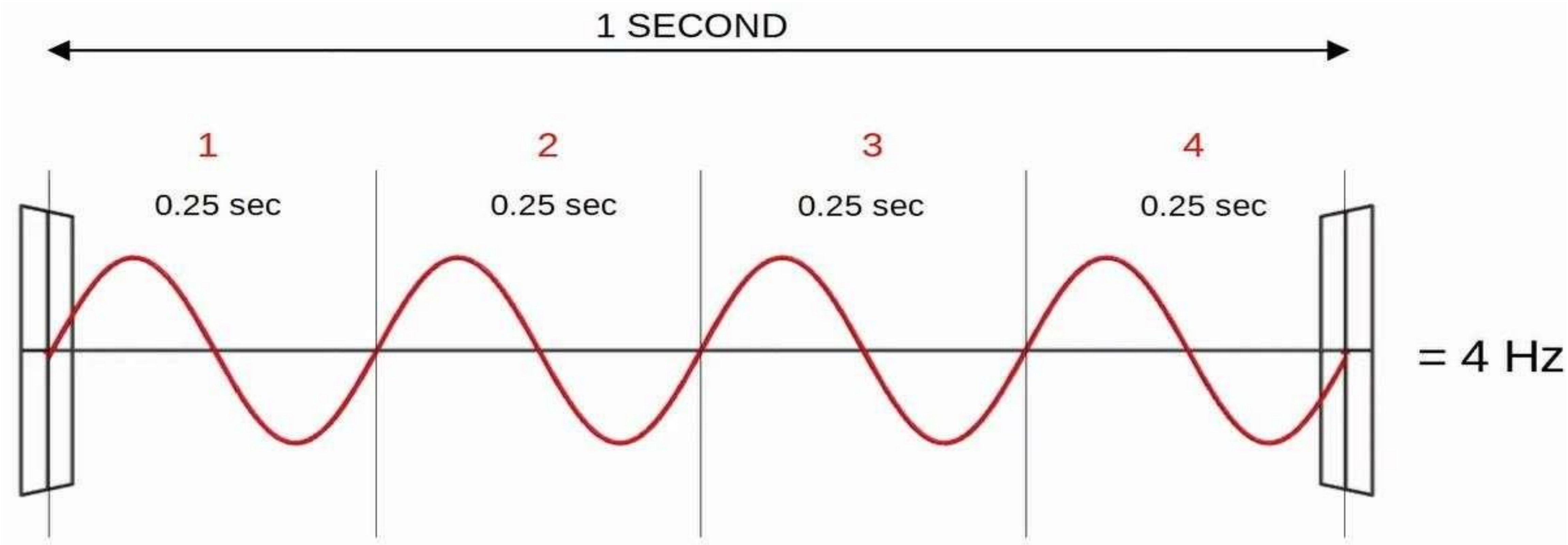
Radio Frequency-

Radia frequency Band-

- 2.4 GHz Band Range- 2.400 GHz to 2.48 35 GHz
- 5 GHz Band, Range- 5.150 GHz to 5.825 GHz



- Another important term is **period**, the amount of time of one cycle.
  → If the **frequency** is 4 Hz, the **period** is 0.25 seconds.

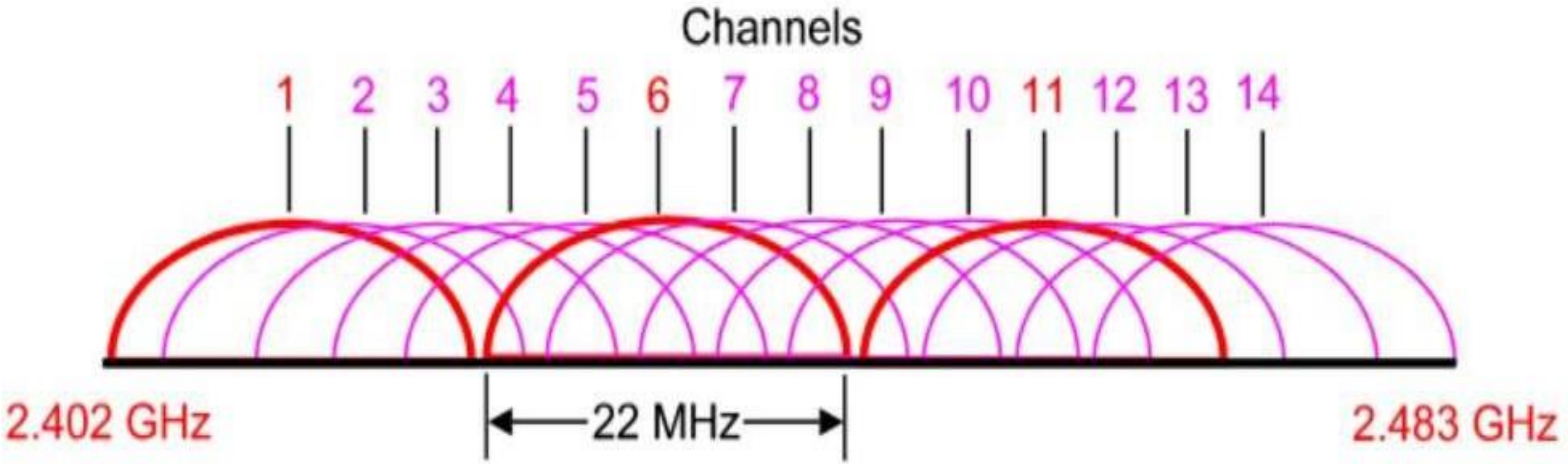
Channels

- Each band is divided up into multiple "Channel"
- Devices are configured to transmit and receive traffic on one (or more) of these channels.

Channel	F_0 (MHz)	Frequency Range (MHz)	North America[8]	Japan [8]	Most of world [8][9][10][11] [12][13][14][15]
1	2412	2401–2423	Yes	Yes	Yes
2	2417	2406–2428	Yes	Yes	Yes
3	2422	2411–2433	Yes	Yes	Yes
4	2427	2416–2438	Yes	Yes	Yes
5	2432	2421–2443	Yes	Yes	Yes

- In Small wireless LAN area within one AP (access Point) use any channel.
- In large wireless LAN are, it's important to use no overlapping channels.
- In 2.4 GHz band, it is highly recommended to use these channels.
- Channel- **1,6,11**

Channels



Wireless Standards

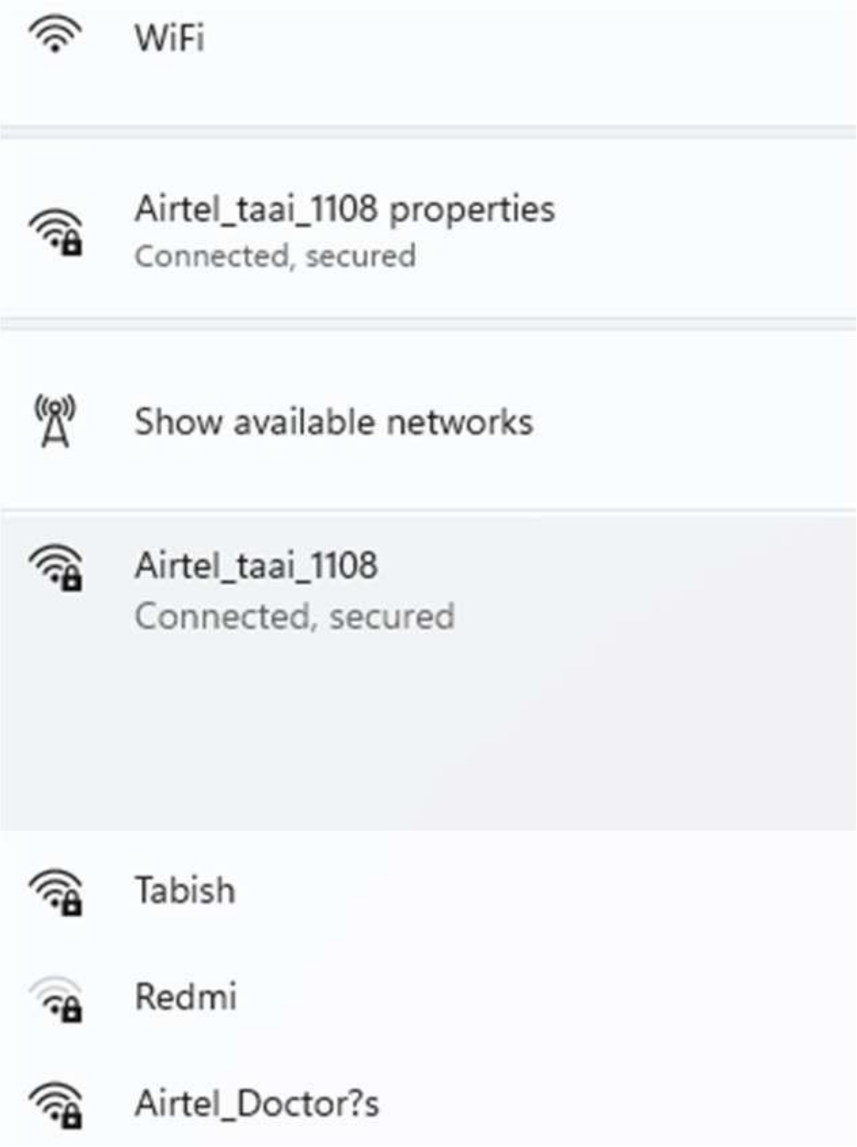
IEEE Standard	Frequency/Medium	Speed	Transmission Range
802.11	2.4GHz RF	1 to 2Mbps	20 feet indoors.
802.11A	5GHz	Up to 54Mbps	25 to 75 feet indoors; range can be affected by building materials.
802.11B	2.4GHz	Up to 11Mbps	Up to 150 feet indoors; range can be affected by building materials.
802.11 G	2.4GHz	Up to 54Mbps	Up to 150 feet indoors; range can be affected by building materials.
802.11 N	2.4GHz/5GHz	Up to 600Mbps	175+ feet indoors; range can be affected by building materials.

Radio Frequency-

SSID (Service set identifier)

- This is name to WIFI
- This is human readable name, which identify the service

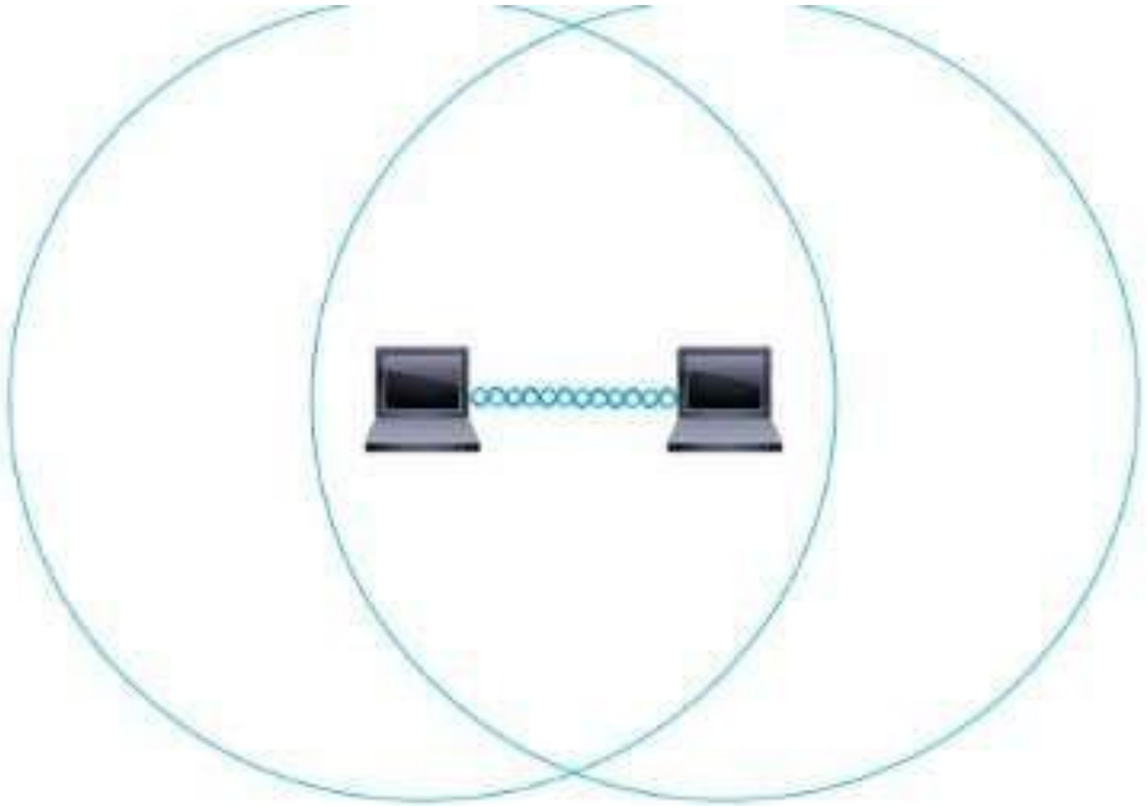
- Its 32 Unique ID character.
- SSID can be hide and visible
- SSID is broadcast by default



Radio Frequency-

IBBS (Identifier Basic service set)

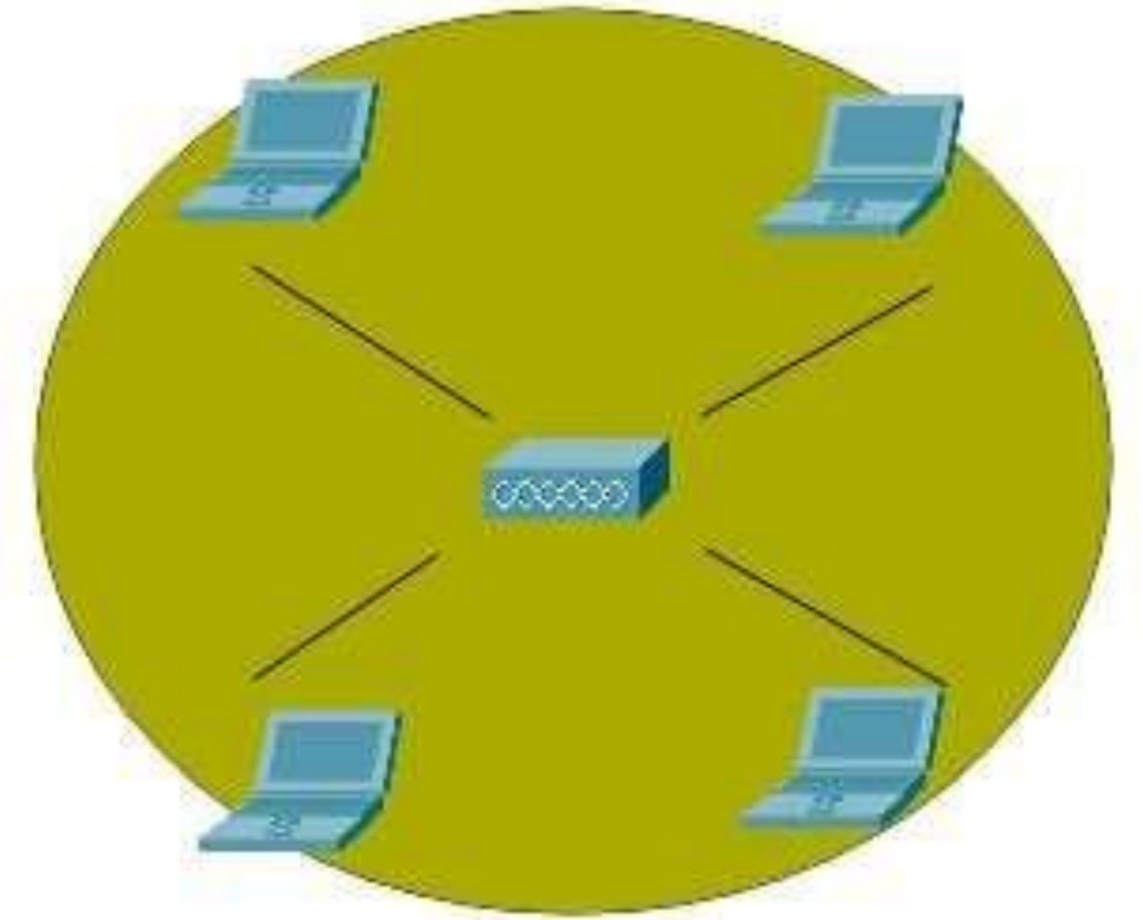
- An IBBS is a wireless network in which two or more wireless devices connected directly without an AP (access point)



- Also called **ad hoc network.**
- It is used to share files between wireless devices.
- No scalable in beyond few devices.

BSS (Basic Service set)

A BSS is a kind of infrastructure service set in which client connect to each other via an AP (access point)

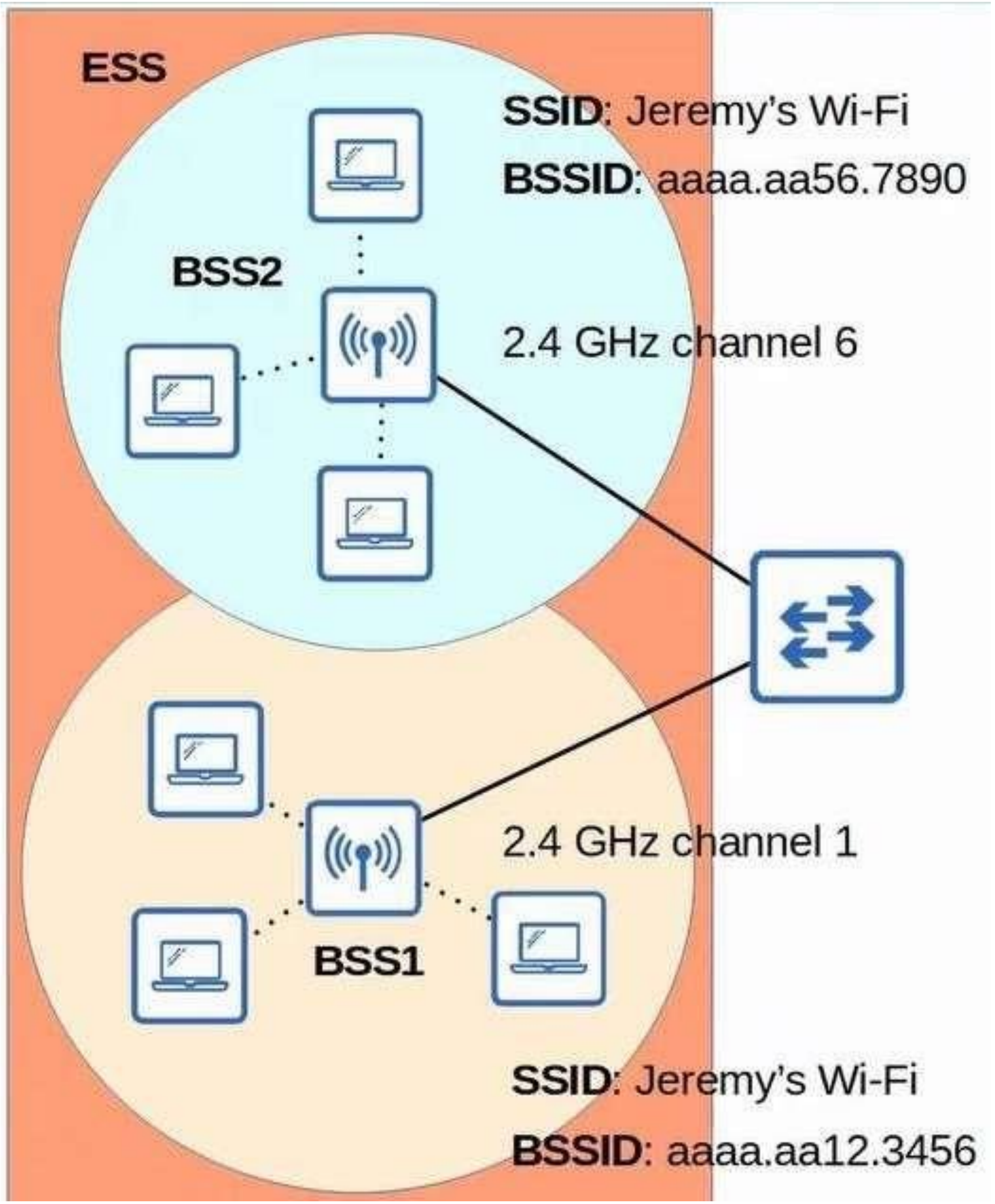


BSA (Basic service area)

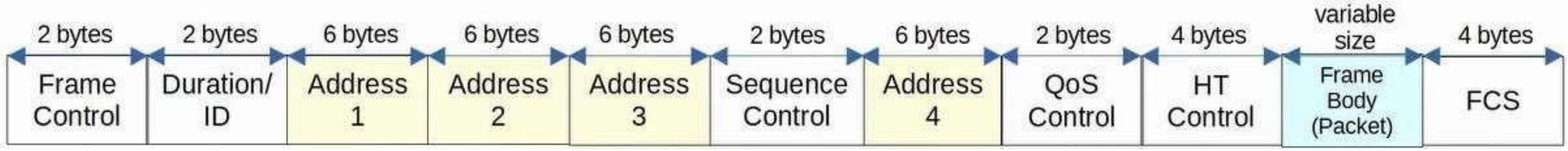
The area around AP is called **BSA.** .

ESS (Extended service set)

- To create large wireless LANs beyond the range of single AP, we use ESS.
- APs with their own BSSs are connected by a wired Network
- Each BSS uses the same SSID
- Each BSS has a unique BSSID



802.11 Frame format- 802.11 frames have a different format the 802.3 ethernet frames.



- Frame control- Provides information such as the message type and subtype
- Duration/ID- Depending on the message type, this field can indicate- in time (in microseconds) the channel will be dedicated for transmission of the frame.
- Address: up to four can be present in an 802.11 frame, which address are present and their order, depends on the message type
- Destination address (DA)- Final recipient of the frame
- Source address (SA)- Original sender of the frame
- Receiver address (RA)- Immediate receive of the frame
- Transmitter address (TA)- Immediate sender of the frame
- Sequence control- Used to reassemble fragments and eliminate duplicate frames
- Qos (Quality of service)- Used to prioritize traffic
- High through put control (HT)- Added in 802.11n to enable high through put operations
- Frame Body packet- This is variable Data size
- FCS (Frame check sequence) used to check error.

802.11 message types-

There are three 802.11 messages types

1 Management- Used to manage BSS (Basic Service set)

- Beacon
- Probe request, probe response
- Authentication request, authentication response

2- Control- Used to control access to the medium (radio frequency).

Assist with delivery of management and data frames

- RTS (Request and send)
- CTS Clear to send)
- ACK

3. Data- Used to send actual data

Access Point Types-

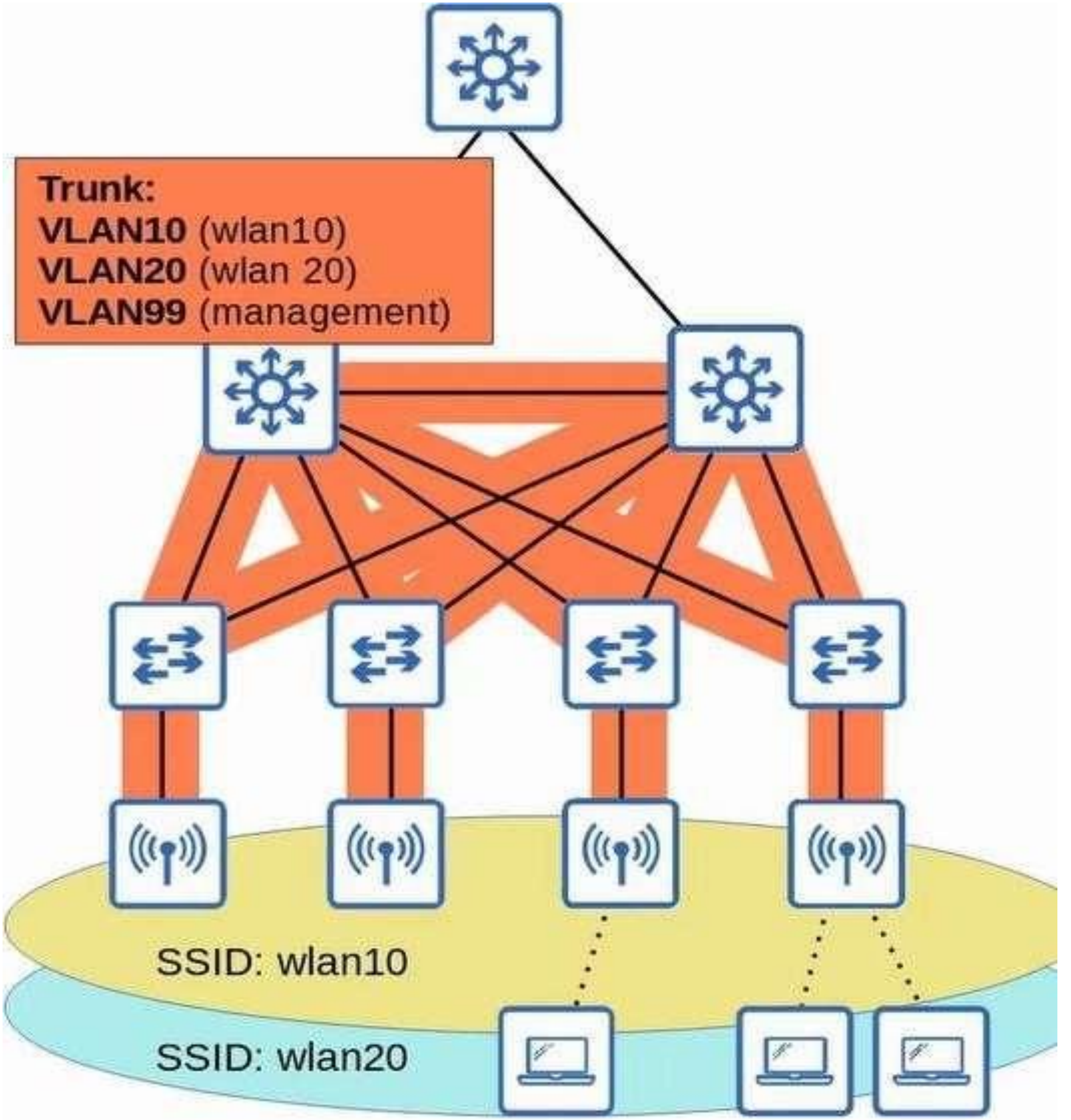
There are three types of APs-

1 - Autonomous Access Point (Aps)
2 - Lightweight access Point
3 - Cloud Based Access Point

1. Autonomous Access Point-

- This is self-contained system don’t rely on WLC (wireless access point) –
- Autonomous APs are independent, no need WLC to configure it.
- Autonomous APs are configured individually
- Autonomous APs are use in small network, where less then 5 APs.
- This AP can be access by console, Telnet/SSH and HTTPs web browser
- An IP address need for remote management
- APs are connected to switches with Trunk link, because of Two SSID or Two VLAN

1. Autonomous Access Point-



Lightweight Access Point (LAP)-
- The function of access point can be split between the AP and WLC.
- Lightweight APs handle real-time operation like, transmitting/receiving, RF traffic, encryption, decryption of the traffic sending out beacon, probe etc
- Other Function of WLC example- RF management, security, client association/roaming management etc
- This is called Split-MAC architecture.
- The WLC (wireless LAN controller) is also used to centrally configure the lightweight APs.
- The WLC and lightweight APs use a protocol called CAPWAP (control and provisioning of wireless) to communicate
- This protocol based on LWAPP (lightweight access point protocol)
- Two tunnels are created between each AP and WLC,
- Control tunnel- This tunnel used to configure APs and control/manage the operation. All traffic in the tunnel encrypted. UDP Port 5246
- Data Tunnel- This tunnel for all traffic from wireless client is sent to through tunnel to WLC.
- Traffic doesn’t go directly to the wired network. Traffic is not encrypted by default.
- No need for trunk, because traffic go through Tunnel, even access point connect to switch access port.

Lightweight Access Point (LAP)-